Lab 8

Question 1

Practice 1

```
frank@cis106:~$ cat /etc/passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
list:x:38:38:Mailing List Manager:/var/list:/usr/sbin/nologin
irc:x:39:39:ircd:/run/ircd:/usr/sbin/nologin
_apt:x:42:65534::/nonexistent:/usr/sbin/nologin
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
systemd-network:x:998:998:systemd Network Management:/:/usr/sbin/nologin
dhcpcd:x:100:65534:DHCP Client Daemon:/usr/lib/dhcpcd:/bin/false
tss:x:101:103:TPM software stack:/var/lib/tpm:/bin/false
systemd-timesync:x:991:991:systemd Time Synchronization:/:/usr/sbin/nologin
messagebus:x:990:990:System Message Bus:/nonexistent:/usr/sbin/nologin
dnsmasq:x:999:65534:dnsmasq:/var/lib/misc:/usr/sbin/nologin
avahi:x:102:107:Avahi mDNS daemon:/run/avahi-daemon:/usr/sbin/nologin
speech-dispatcher:x:103:29:Speech Dispatcher:/run/speech-dispatcher:/bin/false
usbmux:x:104:46:usbmux daemon:/var/lib/usbmux:/usr/sbin/nologin
cups-pk-helper:x:105:108:user for cups-pk-helper service:/nonexistent:/usr/sbin/nologin
fwupd-refresh:x:989:989:Firmware update daemon:/var/lib/fwupd:/usr/sbin/nologin
geoclue:x:106:110::/var/lib/geoclue:/usr/sbin/nologin
gnome-remote-desktop:x:988:988:GNOME Remote Desktop:/var/lib/gnome-remote-desktop:/usr/sbin/nologin
saned:x:107:111::/var/lib/saned:/usr/sbin/nologin
polkitd:x:987:987:User for polkitd:/:/usr/sbin/nologin
rtkit:x:108:112:RealtimeKit:/proc:/usr/sbin/nologin
colord:x:109:113:colord colour management daemon:/var/lib/colord:/usr/sbin/nologin
Debian-gdm:x:110:114:Gnome Display Manager:/var/lib/gdm3:/bin/false
frank:x:1000:1000:frank,,,:/home/frank:/bin/bash
_flatpak:x:111:115:Flatpak system-wide installation helper:/nonexistent:/usr/sbin/nologin
vboxadd:x:997:1::/var/run/vboxadd:/bin/false
frank@cis106:~$
```

```
frank@cis106:~$ tac /etc/passwd
vboxadd:x:997:1::/var/run/vboxadd:/bin/false
_flatpak:x:111:115:Flatpak system-wide installation helper:/nonexistent:/usr/sbin/nologin
frank:x:1000:1000:frank,,,:/home/frank:/bin/bash
Debian-gdm:x:110:114:Gnome Display Manager:/var/lib/gdm3:/bin/false
colord:x:109:113:colord colour management daemon:/var/lib/colord:/usr/sbin/nologin
rtkit:x:108:112:RealtimeKit:/proc:/usr/sbin/nologin
polkitd:x:987:987:User for polkitd:/:/usr/sbin/nologin
saned:x:107:111::/var/lib/saned:/usr/sbin/nologin
gnome-remote-desktop:x:988:988:GNOME Remote Desktop:/var/lib/gnome-remote-desktop:/usr/sbin/nologin
geoclue:x:106:110::/var/lib/geoclue:/usr/sbin/nologin
fwupd-refresh:x:989:989:Firmware update daemon:/var/lib/fwupd:/usr/sbin/nologin
cups-pk-helper:x:105:108:user for cups-pk-helper service:/nonexistent:/usr/sbin/nologin
usbmux:x:104:46:usbmux daemon:/var/lib/usbmux:/usr/sbin/nologin
speech-dispatcher:x:103:29:Speech Dispatcher:/run/speech-dispatcher:/bin/false
avahi:x:102:107:Avahi mDNS daemon:/run/avahi-daemon:/usr/sbin/nologin
dnsmasq:x:999:65534:dnsmasq:/var/lib/misc:/usr/sbin/nologin
messagebus:x:990:990:System Message Bus:/nonexistent:/usr/sbin/nologin
systemd-timesync:x:991:991:systemd Time Synchronization:/:/usr/sbin/nologin
tss:x:101:103:TPM software stack:/var/lib/tpm:/bin/false
dhcpcd:x:100:65534:DHCP Client Daemon:/usr/lib/dhcpcd:/bin/false
systemd-network:x:998:998:systemd Network Management:/:/usr/sbin/nologin
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
_apt:x:42:65534::/nonexistent:/usr/sbin/nologin
irc:x:39:39:ircd:/run/ircd:/usr/sbin/nologin
list:x:38:38:Mailing List Manager:/var/list:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
games:x:5:60:games:/usr/games:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
sys:x:3:3:sys:/dev:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
root:x:0:0:root:/root:/bin/bash
frank@cis106:~$
```

```
frank@cis106:~$ cat -nE /etc/passwd
     1  root:x:0:0:root:/root:/bin/bash$
     2  daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin$
     3  bin:x:2:2:bin:/bin:/usr/sbin/nologin$
     4  sys:x:3:3:sys:/dev:/usr/sbin/nologin$
     5  sync:x:4:65534:sync:/bin:/bin/sync$
     6  games:x:5:60:games:/usr/games:/usr/sbin/nologin$
     7  man:x:6:12:man:/var/cache/man:/usr/sbin/nologin$
     8  lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin$
     9  mail:x:8:8:mail:/var/mail:/usr/sbin/nologin$
    10  news:x:9:9:news:/var/spool/news:/usr/sbin/nologin$
    11  uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin$
    12  proxy:x:13:13:proxy:/bin:/usr/sbin/nologin$
    13  www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin$
    14  backup:x:34:34:backup:/var/backups:/usr/sbin/nologin$
    15  list:x:38:38:Mailing List Manager:/var/list:/usr/sbin/nologin$
    16  irc:x:39:39:ircd:/run/ircd:/usr/sbin/nologin$
    17  _apt:x:42:65534::/nonexistent:/usr/sbin/nologin$
    18  nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin$
    19  systemd-network:x:998:998:systemd Network Management:/:/usr/sbin/nologin$
    20  dhcpcd:x:100:65534:DHCP Client Daemon:/usr/lib/dhcpcd:/bin/false$
    21  tss:x:101:103:TPM software stack:/var/lib/tpm:/bin/false$
    22  systemd-timesync:x:991:991:systemd Time Synchronization:/:/usr/sbin/nologin$
    23  messagebus:x:990:990:System Message Bus:/nonexistent:/usr/sbin/nologin$
    24  dnsmasq:x:999:65534:dnsmasq:/var/lib/misc:/usr/sbin/nologin$
    25  avahi:x:102:107:Avahi mDNS daemon:/run/avahi-daemon:/usr/sbin/nologin$
    26  speech-dispatcher:x:103:29:Speech Dispatcher:/run/speech-dispatcher:/bin/false$
    27  usbmux:x:104:46:usbmux daemon:/var/lib/usbmux:/usr/sbin/nologin$
    28  cups-pk-helper:x:105:108:user for cups-pk-helper service:/nonexistent:/usr/sbin/nologin$
    29  fwupd-refresh:x:989:989:Firmware update daemon:/var/lib/fwupd:/usr/sbin/nologin$
    30  geoclue:x:106:110::/var/lib/geoclue:/usr/sbin/nologin$
    31  gnome-remote-desktop:x:988:988:GNOME Remote Desktop:/var/lib/gnome-remote-desktop:/usr/sbin/nologin$
    32  saned:x:107:111::/var/lib/saned:/usr/sbin/nologin$
    33  polkitd:x:987:987:User for polkitd:/:/usr/sbin/nologin$
    34  rtkit:x:108:112:RealtimeKit:/proc:/usr/sbin/nologin$
    35  colord:x:109:113:colord colour management daemon:/var/lib/colord:/usr/sbin/nologin$
    36  Debian-gdm:x:110:114:Gnome Display Manager:/var/lib/gdm3:/bin/false$
    37  frank:x:1000:1000:frank,,,:/home/frank:/bin/bash$
    38  _flatpak:x:111:115:Flatpak system-wide installation helper:/nonexistent:/usr/sbin/nologin$
    39  vboxadd:x:997:1::/var/run/vboxadd:/bin/false$
frank@cis106:~$
```

```
    16  irc:x:39:39:ircd:/run/ircd:/usr/sbin/nologin$
    17  _apt:x:42:65534::/nonexistent:/usr/sbin/nologin$
    18  nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin$
    19  systemd-network:x:998:998:systemd Network Management:/:/usr/sbin/nologin$
    20  dhcpcd:x:100:65534:DHCP Client Daemon:/usr/lib/dhcpcd:/bin/false$
    21  tss:x:101:103:TPM software stack:/var/lib/tpm:/bin/false$
    22  systemd-timesync:x:991:991:systemd Time Synchronization:/:/usr/sbin/nologin$
    23  messagebus:x:990:990:System Message Bus:/nonexistent:/usr/sbin/nologin$
    24  dnsmasq:x:999:65534:dnsmasq:/var/lib/misc:/usr/sbin/nologin$
    25  avahi:x:102:107:Avahi mDNS daemon:/run/avahi-daemon:/usr/sbin/nologin$
    26  speech-dispatcher:x:103:29:Speech Dispatcher:/run/speech-dispatcher:/bin/false$
    27  usbmux:x:104:46:usbmux daemon:/var/lib/usbmux:/usr/sbin/nologin$
    28  cups-pk-helper:x:105:108:user for cups-pk-helper service:/nonexistent:/usr/sbin/nologin$
    29  fwupd-refresh:x:989:989:Firmware update daemon:/var/lib/fwupd:/usr/sbin/nologin$
    30  geoclue:x:106:110::/var/lib/geoclue:/usr/sbin/nologin$
    31  gnome-remote-desktop:x:988:988:GNOME Remote Desktop:/var/lib/gnome-remote-desktop:/usr/sbin/nologin$
    32  saned:x:107:111::/var/lib/saned:/usr/sbin/nologin$
    33  polkitd:x:987:987:User for polkitd:/:/usr/sbin/nologin$
    34  rtkit:x:108:112:RealtimeKit:/proc:/usr/sbin/nologin$
    35  colord:x:109:113:colord colour management daemon:/var/lib/colord:/usr/sbin/nologin$
    36  Debian-gdm:x:110:114:Gnome Display Manager:/var/lib/gdm3:/bin/false$
    37  frank:x:1000:1000:frank,,,:/home/frank:/bin/bash$
    38  _flatpak:x:111:115:Flatpak system-wide installation helper:/nonexistent:/usr/sbin/nologin$
    39  vboxadd:x:997:1::/var/run/vboxadd:/bin/false$
frank@cis106:~$ head -n 5 /etc/passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
frank@cis106:~$ tail -n 5 /etc/passwd
colord:x:109:113:colord colour management daemon:/var/lib/colord:/usr/sbin/nologin
Debian-gdm:x:110:114:Gnome Display Manager:/var/lib/gdm3:/bin/false
frank:x:1000:1000:frank,,,:/home/frank:/bin/bash
_flatpak:x:111:115:Flatpak system-wide installation helper:/nonexistent:/usr/sbin/nologin
vboxadd:x:997:1::/var/run/vboxadd:/bin/false
frank@cis106:~$ tail -1 /etc/passwd
vboxadd:x:997:1::/var/run/vboxadd:/bin/false
frank@cis106:~$ head -1 /etc/passwd
root:x:0:0:root:/root:/bin/bash
frank@cis106:~$
```

```
Tilix: frank@cis106: ~
1: frank@cis106: ~
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15842  Archive Foundation, how to help produce our new eBooks, and how to
15843  subscribe to our email newsletter to hear about new eBooks.
15844
15845
frank@cis106:~$
```

Practice 2

```
frank@cis106:~$ cut -d":" -f1 /etc/passwd
root
daemon
bin
sys
sync
games
man
lp
mail
news
uucp
proxy
www-data
backup
list
irc
_apt
nobody
systemd-network
dhcpcd
tss
systemd-timesync
messagebus
dnsmasq
avahi
speech-dispatcher
usbmux
cups-pk-helper
fwupd-refresh
geoclue
gnome-remote-desktop
saned
polkitd
rtkit
colord
Debian-gdm
frank
_flatpak
vboxadd
frank@cis106:~$
```

```
frank@cis106:~$ cut -d":" -f1,6 /etc/passwd
root:/root
daemon:/usr/sbin
bin:/bin
sys:/dev
sync:/bin
games:/usr/games
man:/var/cache/man
lp:/var/spool/lpd
mail:/var/mail
news:/var/spool/news
uucp:/var/spool/uucp
proxy:/bin
www-data:/var/www
backup:/var/backups
list:/var/list
irc:/run/ircd
_apt:/nonexistent
nobody:/nonexistent
systemd-network:/
dhcpcd:/usr/lib/dhcpcd
tss:/var/lib/tpm
systemd-timesync:/
messagebus:/nonexistent
dnsmasq:/var/lib/misc
avahi:/run/avahi-daemon
speech-dispatcher:/run/speech-dispatcher
usbmux:/var/lib/usbmux
cups-pk-helper:/nonexistent
fwupd-refresh:/var/lib/fwupd
geoclue:/var/lib/geoclue
gnome-remote-desktop:/var/lib/gnome-remote-desktop
saned:/var/lib/saned
polkitd:/
rtkit:/proc
colord:/var/lib/colord
Debian-gdm:/var/lib/gdm3
frank:/home/frank
_flatpak:/nonexistent
vboxadd:/var/run/vboxadd
frank@cis106:~$
```

```
frank@cis106:~$ cut -d":" -f1,6,7 --output-delimiter=" > " /etc/passwd
root > /root > /bin/bash
daemon > /usr/sbin > /usr/sbin/nologin
bin > /bin > /usr/sbin/nologin
sys > /dev > /usr/sbin/nologin
sync > /bin > /bin/sync
games > /usr/games > /usr/sbin/nologin
man > /var/cache/man > /usr/sbin/nologin
lp > /var/spool/lpd > /usr/sbin/nologin
mail > /var/mail > /usr/sbin/nologin
news > /var/spool/news > /usr/sbin/nologin
uucp > /var/spool/uucp > /usr/sbin/nologin
proxy > /bin > /usr/sbin/nologin
www-data > /var/www > /usr/sbin/nologin
backup > /var/backups > /usr/sbin/nologin
list > /var/list > /usr/sbin/nologin
irc > /run/ircd > /usr/sbin/nologin
_apt > /nonexistent > /usr/sbin/nologin
nobody > /nonexistent > /usr/sbin/nologin
systemd-network > / > /usr/sbin/nologin
dhcpcd > /usr/lib/dhcpcd > /bin/false
tss > /var/lib/tpm > /bin/false
systemd-timesync > / > /usr/sbin/nologin
messagebus > /nonexistent > /usr/sbin/nologin
dnsmasq > /var/lib/misc > /usr/sbin/nologin
avahi > /run/avahi-daemon > /usr/sbin/nologin
speech-dispatcher > /run/speech-dispatcher > /bin/false
usbmux > /var/lib/usbmux > /usr/sbin/nologin
cups-pk-helper > /nonexistent > /usr/sbin/nologin
fwupd-refresh > /var/lib/fwupd > /usr/sbin/nologin
geoclue > /var/lib/geoclue > /usr/sbin/nologin
gnome-remote-desktop > /var/lib/gnome-remote-desktop > /usr/sbin/nologin
saned > /var/lib/saned > /usr/sbin/nologin
polkitd > / > /usr/sbin/nologin
rtkit > /proc > /usr/sbin/nologin
colord > /var/lib/colord > /usr/sbin/nologin
Debian-gdm > /var/lib/gdm3 > /bin/false
frank > /home/frank > /bin/bash
_flatpak > /nonexistent > /usr/sbin/nologin
vboxadd > /var/run/vboxadd > /bin/false
frank@cis106:~$
```

```
Debian-gdm > /var/lib/gdm3 > /bin/false
frank > /home/frank > /bin/bash
_flatpak > /nonexistent > /usr/sbin/nologin
vboxadd > /var/run/vboxadd > /bin/false
frank@cis106:~$ cut -d";" -f1 ~/Documents/sample_files/Csv/cereal.csv
name
String
100% Bran
100% Natural Bran
All-Bran
All-Bran with Extra Fiber
Almond Delight
Apple Cinnamon Cheerios
Apple Jacks
Basic 4
Bran Chex
Bran Flakes
Cap'n'Crunch
Cheerios
Cinnamon Toast Crunch
Clusters
Cocoa Puffs
Corn Chex
Corn Flakes
Corn Pops
Count Chocula
Cracklin' Oat Bran
Cream of Wheat (Quick)
Crispix
Crispy Wheat & Raisins
Double Chex
Froot Loops
Frosted Flakes
Frosted Mini-Wheats
Fruit & Fibre Dates, Walnuts, and Oats
Fruitful Bran
Fruity Pebbles
Golden Crisp
Golden Grahams
Grape Nuts Flakes
Grape-Nuts
Great Grains Pecan
Honey Graham Ohs
Honey Nut Cheerios
Honey-comb
Just Right Crunchy  Nuggets
Just Right Fruit & Nut
Kix
Life
Lucky Charms
Maypo
```

```
Trix
Wheat Chex
Wheaties
Wheaties Honey Gold
frank@cis106:~$ cut -d";" -f1,4 --output-delimiter=" calories = " ~/Documents/sample_files/Csv/cereal.csv
name calories = calories
String calories = Int
100% Bran calories = 70
100% Natural Bran calories = 120
All-Bran calories = 70
All-Bran with Extra Fiber calories = 50
Almond Delight calories = 110
Apple Cinnamon Cheerios calories = 110
Apple Jacks calories = 110
Basic 4 calories = 130
Bran Chex calories = 90
Bran Flakes calories = 90
Cap'n'Crunch calories = 120
Cheerios calories = 110
Cinnamon Toast Crunch calories = 120
Clusters calories = 110
Cocoa Puffs calories = 110
Corn Chex calories = 110
Corn Flakes calories = 100
Corn Pops calories = 110
Count Chocula calories = 110
Cracklin' Oat Bran calories = 110
Cream of Wheat (Quick) calories = 100
Crispix calories = 110
Crispy Wheat & Raisins calories = 100
Double Chex calories = 100
Froot Loops calories = 110
Frosted Flakes calories = 110
Frosted Mini-Wheats calories = 100
Fruit & Fibre Dates, Walnuts, and Oats calories = 120
Fruitful Bran calories = 120
Fruity Pebbles calories = 110
Golden Crisp calories = 100
Golden Grahams calories = 110
Grape Nuts Flakes calories = 100
Grape-Nuts calories = 110
Great Grains Pecan calories = 120
Honey Graham Ohs calories = 120
Honey Nut Cheerios calories = 110
Honey-comb calories = 110
Just Right Crunchy  Nuggets calories = 110
Just Right Fruit & Nut calories = 140
Kix calories = 110
Life calories = 100
Lucky Charms calories = 110
Maypo calories = 100
```

Practice 3

```
Tilix: frank@cis106: ~/Documents/sample_files/Csv
1: frank@cis106: ~/Documents/sample_files/Csv
frank@cis106:~$ cd Documents/sample_files/Csv/
frank@cis106:~/Documents/sample_files/Csv$ sort cars.csv
AMC Ambassador Brougham;13.0;8;360.0;175.0;3821.;11.0;73;US
AMC Ambassador DPL;15.0;8;390.0;190.0;3850.;8.5;70;US
AMC Ambassador SST;17.0;8;304.0;150.0;3672.;11.5;72;US
AMC Concord;19.4;6;232.0;90.00;3210.;17.2;78;US
AMC Concord;24.3;4;151.0;90.00;3003.;20.1;80;US
AMC Concord d/l;18.1;6;258.0;120.0;3410.;15.1;78;US
AMC Concord DL;23.0;4;151.0;0;3035.;20.5;82;US
AMC Concord DL 6;20.2;6;232.0;90.00;3265.;18.2;79;US
AMC Gremlin;18.0;6;232.0;100.0;2789.;15.0;73;US
AMC Gremlin;19.0;6;232.0;100.0;2634.;13.0;71;US
AMC Gremlin;20.0;6;232.0;100.0;2914.;16.0;75;US
AMC Gremlin;21.0;6;199.0;90.00;2648.;15.0;70;US
AMC Hornet;18.0;6;199.0;97.00;2774.;15.5;70;US
AMC Hornet;18.0;6;232.0;100.0;2945.;16.0;73;US
AMC Hornet;19.0;6;232.0;100.0;2901.;16.0;74;US
AMC Hornet;22.5;6;232.0;90.00;3085.;17.6;76;US
AMC Hornet Sportabout (sw);18.0;6;258.0;110.0;2962.;13.5;71;US
AMC Matador;14.0;8;304.0;150.0;3672.;11.5;73;US
AMC Matador;15.0;6;258.0;110.0;3730.;19.0;75;US
AMC Matador;15.5;8;304.0;120.0;3962.;13.9;76;US
AMC Matador;16.0;6;258.0;110.0;3632.;18.0;74;US
AMC Matador;18.0;6;232.0;100.0;3288.;15.5;71;US
AMC Matador (sw);14.0;8;304.0;150.0;4257.;15.5;74;US
AMC Matador (sw);15.0;8;304.0;150.0;3892.;12.5;72;US
AMC Pacer;19.0;6;232.0;90.00;3211.;17.0;75;US
AMC Pacer d/l;17.5;6;258.0;95.00;3193.;17.8;76;US
AMC Rebel SST;16.0;8;304.0;150.0;3433.;12.0;70;US
AMC Rebel SST (sw);0;8;360.0;175.0;3850.;11.0;70;US
AMC Spirit DL;27.4;4;121.0;80.00;2670.;15.0;79;US
Audi 100LS;20.0;4;114.0;91.00;2582.;14.0;73;Europe
Audi 100LS;23.0;4;115.0;95.00;2694.;15.0;75;Europe
Audi 100 LS;24.0;4;107.0;90.00;2430.;14.5;70;Europe
Audi 4000;34.3;4;97.00;78.00;2188.;15.8;80;Europe
Audi 5000;20.3;5;131.0;103.0;2830.;15.9;78;Europe
Audi 5000s (diesel);36.4;5;121.0;67.00;2950.;19.9;80;Europe
Audi Fox;29.0;4;98.00;83.00;2219.;16.5;74;Europe
BMW 2002;26.0;4;121.0;113.0;2234.;12.5;70;Europe
BMW 320i;21.5;4;121.0;110.0;2600.;12.8;77;Europe
Buick Century;17.0;6;231.0;110.0;3907.;21.0;75;US
Buick Century;22.4;6;231.0;110.0;3415.;15.8;81;US
```

```
Tilix: frank@cis106: ~/Documents/sample_files/Csv
1: frank@cis106: ~/Documents/sample_files/Csv
Toyota Cressida;25.4;6;168.0;116.0;2900.;12.6;81;Japan
Toyota Mark II;19.0;6;156.0;108.0;2930.;15.5;76;Japan
Toyota Mark II;20.0;6;156.0;122.0;2807.;13.5;73;Japan
Toyota Starlet;39.1;4;79.00;58.00;1755.;16.9;81;Japan
Toyota Tercel;37.7;4;89.00;62.00;2050.;17.3;81;Japan
Triumph TR7 Coupe;35.0;4;122.0;88.00;2500.;15.1;80;Europe
Volkswagen 1131 Deluxe Sedan;26.0;4;97.00;46.00;1835.;20.5;70;Europe
Volkswagen 411 (sw);22.0;4;121.0;76.00;2511.;18.0;72;Europe
Volkswagen Dasher;25.0;4;90.00;71.00;2223.;16.5;75;Europe
Volkswagen Dasher;26.0;4;79.00;67.00;1963.;15.5;74;Europe
Volkswagen Dasher;30.5;4;97.00;78.00;2190.;14.1;77;Europe
Volkswagen Dasher (diesel);43.4;4;90.00;48.00;2335.;23.7;80;Europe
Volkswagen Jetta;33.0;4;105.0;74.00;2190.;14.2;81;Europe
Volkswagen Model 111;27.0;4;97.00;60.00;1834.;19.0;71;Europe
Volkswagen Pickup;44.0;4;97.00;52.00;2130.;24.6;82;Europe
Volkswagen Rabbit;29.0;4;90.00;70.00;1937.;14.0;75;Europe
Volkswagen Rabbit;29.0;4;90.00;70.00;1937.;14.2;76;Europe
Volkswagen Rabbit;29.5;4;97.00;71.00;1825.;12.2;76;Europe
Volkswagen Rabbit;29.8;4;89.00;62.00;1845.;15.3;80;Europe
Volkswagen Rabbit;41.5;4;98.00;76.00;2144.;14.7;80;Europe
Volkswagen Rabbit C (Diesel);44.3;4;90.00;48.00;2085.;21.7;80;Europe
Volkswagen Rabbit Custom;29.0;4;97.00;78.00;1940.;14.5;77;Europe
Volkswagen Rabbit Custom;31.9;4;89.00;71.00;1925.;14.0;79;Europe
Volkswagen Rabbit Custom Diesel;43.1;4;90.00;48.00;1985.;21.5;78;Europe
Volkswagen Rabbit l;36.0;4;105.0;74.00;1980.;15.3;82;Europe
Volkswagen Scirocco;31.5;4;89.00;71.00;1990.;14.9;78;Europe
Volkswagen Super Beetle 117;0;4;97.00;48.00;1978.;20.0;71;Europe
Volkswagen Super Beetle;26.0;4;97.00;46.00;1950.;21.0;73;Europe
Volkswagen Type 3;23.0;4;97.00;54.00;2254.;23.5;72;Europe
Volvo 144ea;19.0;4;121.0;112.0;2868.;15.5;73;Europe
Volvo 145e (sw);18.0;4;121.0;112.0;2933.;14.5;72;Europe
Volvo 244DL;22.0;4;121.0;98.00;2945.;14.5;75;Europe
Volvo 245;20.0;4;130.0;102.0;3150.;15.7;76;Europe
Volvo 264gl;17.0;6;163.0;125.0;3140.;13.6;78;Europe
Volvo Diesel;30.7;6;145.0;76.00;3160.;19.6;81;Europe
frank@cis106:~/Documents/sample_files/Csv$ sort -o sorted_cars.csv cars.csv
frank@cis106:~/Documents/sample_files/Csv$ wc -l sorted_cars.csv
408 sorted_cars.csv
frank@cis106:~/Documents/sample_files/Csv$ wc -m sorted_cars.csv
22663 sorted_cars.csv
frank@cis106:~/Documents/sample_files/Csv$
```

Question 2

```
Tilix: frank@cis106: ~
1: frank@cis106: ~
    61  2025-10-04 01:57:16 Walter /dashboard FAIL
    62  2025-09-01 22:22:06 Faye /download DENIED
    63  2025-03-08 21:10:30 Rize /dashboard SUCCESS
    64  2025-08-16 07:20:13 Rize /logout SUCCESS
    65  2025-06-08 17:28:21 Rize /upload FAIL
    66  2025-04-23 16:03:34 Touka /admin FAIL
    67  2025-05-26 20:43:46 Edward /404 DENIED
    68  2025-04-08 20:51:04 Kei /download FAIL
    69  2025-05-22 13:19:58 Haruka /logout SUCCESS
    70  2025-10-09 15:46:36 Honami /admin SUCCESS
    71  2025-06-24 02:13:00 Julia /admin SUCCESS
    72  2025-10-08 19:46:22 Shinji /settings FAIL
    73  2025-10-01 19:35:19 Annie /download SUCCESS
    74  2025-08-20 06:14:20 Ken /logout SUCCESS
    75  2025-05-21 16:53:20 Touka /admin SUCCESS
    76  2025-09-12 15:08:18 Touka /download SUCCESS
    77  2025-12-02 08:07:32 Touka /logout FAIL
    78  2025-08-21 21:25:07 Faye /admin SUCCESS
    79  2025-03-25 21:20:59 Haruka /admin SUCCESS
    80  2025-03-16 21:09:43 Spike /login FAIL
    81  2025-02-27 15:09:11 Seras /upload DENIED
    82  2025-05-27 20:43:15 Rize /download SUCCESS
    83  2025-08-04 18:36:17 Jet /reset DENIED
    84  2025-02-10 21:50:35 Seras /reset SUCCESS
    85  2025-07-21 20:38:39 Rei /upload FAIL
    86  2025-01-10 08:04:17 Asimov /login FAIL
    87  2025-01-04 22:32:01 Faye /logout FAIL
    88  2025-06-20 04:28:48 Annie /dashboard SUCCESS
    89  2025-06-28 16:05:02 Eren /upload FAIL
    90  2025-06-24 08:20:03 Eren /logout SUCCESS
    91  2025-01-12 05:45:12 Shinji /admin DENIED
    92  2025-05-04 18:53:22 Shinji /settings SUCCESS
    93  2025-08-09 09:43:12 Walter /reset FAIL
    94  2025-12-14 21:48:13 Ken /download FAIL
    95  2025-06-25 23:54:08 Arisu /download DENIED
    96  2025-01-18 22:23:34 Spike /settings SUCCESS
    97  2025-12-23 15:18:24 Julia /reset SUCCESS
    98  2025-02-25 11:05:38 Rize /admin DENIED
    99  2025-03-12 15:23:38 Eren /upload FAIL
   100  2025-03-08 13:07:23 Rei /download FAIL
frank@cis106:~$
```

```
Tilix: frank@cis106: ~
1: frank@cis106: ~
    97  2025-12-23 15:18:24 Julia /reset SUCCESS
    98  2025-02-25 11:05:38 Rize /admin DENIED
    99  2025-03-12 15:23:38 Eren /upload FAIL
   100  2025-03-08 13:07:23 Rei /download FAIL
frank@cis106:~$ head -1 access.log
2025-01-03 17:09:02 Ken /dashboard FAIL
frank@cis106:~$ tail -1 access.log
2025-03-08 13:07:23 Rei /download FAIL
frank@cis106:~$ head -n 15 access.log
2025-01-03 17:09:02 Ken /dashboard FAIL
2025-08-09 11:15:28 Edward /reset DENIED
2025-12-18 03:33:07 Jet /settings FAIL
2025-06-04 02:34:36 Honami /admin DENIED
2025-11-26 06:37:54 Rei /settings SUCCESS
2025-07-17 04:35:39 Asimov /logout FAIL
2025-04-23 00:24:48 Asimov /upload FAIL
2025-11-13 22:01:24 Edward /reset FAIL
2025-08-23 20:02:11 Touka /download FAIL
2025-01-26 21:22:50 Rize /admin DENIED
2025-01-22 03:33:11 Honami /logout FAIL
2025-12-04 11:03:31 Jet /logout DENIED
2025-05-12 23:05:02 Shinji /upload DENIED
2025-09-18 15:31:39 Kei /reset DENIED
2025-07-19 22:37:47 Shinji /login FAIL
frank@cis106:~$ tail -n 15 access.log
2025-01-10 08:04:17 Asimov /login FAIL
2025-01-04 22:32:01 Faye /logout FAIL
2025-06-20 04:28:48 Annie /dashboard SUCCESS
2025-06-28 16:05:02 Eren /upload FAIL
2025-06-24 08:20:03 Eren /logout SUCCESS
2025-01-12 05:45:12 Shinji /admin DENIED
2025-05-04 18:53:22 Shinji /settings SUCCESS
2025-08-09 09:43:12 Walter /reset FAIL
2025-12-14 21:48:13 Ken /download FAIL
2025-06-25 23:54:08 Arisu /download DENIED
2025-01-18 22:23:34 Spike /settings SUCCESS
2025-12-23 15:18:24 Julia /reset SUCCESS
2025-02-25 11:05:38 Rize /admin DENIED
2025-03-12 15:23:38 Eren /upload FAIL
2025-03-08 13:07:23 Rei /download FAIL
frank@cis106:~$
```

Tilix: frank@cis106: ~

```
frank@cis106:~$ cut -d " " -f3 access.log
Ken
Edward
Jet
Honami
Rei
Asimov
Asimov
Edward
Touka
Rize
Honami
Jet
Shinji
Kei
Shinji
Kei
Touka
Arisu
Kei
Jet
Rei
Ein
Ein
Julia
Seras
Touka
Seras
Asimov
Walter
Ein
Eren
Faye
Seras
Ken
Edward
Edward
Ken
Asimov
Ein
Faye
Edward
Annie
Annie
Edward
Ken
Ein
Seras
Arisu
Julia
Jet
```

Tilix: frank@cis106: ~

```
frank@cis106:~$ sort -k3 -o sorted_access_log_by_uname.log access.log
frank@cis106:~$ cat sorted_access_log_by_uname.log
2025-09-15 09:51:26 Annie /admin SUCCESS
2025-06-20 04:28:48 Annie /dashboard SUCCESS
2025-04-17 00:27:50 Annie /download SUCCESS
2025-10-01 19:35:19 Annie /download SUCCESS
2025-12-01 06:43:14 Arisu /404 FAIL
2025-06-25 23:54:08 Arisu /download DENIED
2025-03-16 10:27:25 Arisu /reset DENIED
2025-08-06 16:06:52 Arisu /reset DENIED
2025-01-10 08:04:17 Asimov /login FAIL
2025-07-17 04:35:39 Asimov /logout FAIL
2025-09-20 14:12:55 Asimov /logout FAIL
2025-01-14 15:12:14 Asimov /reset FAIL
2025-04-23 00:24:48 Asimov /upload FAIL
2025-05-26 20:43:46 Edward /404 DENIED
2025-07-11 07:04:18 Edward /404 SUCCESS
2025-12-08 21:06:19 Edward /404 SUCCESS
2025-08-27 14:07:23 Edward /admin SUCCESS
2025-12-05 16:20:59 Edward /dashboard SUCCESS
2025-08-09 11:15:28 Edward /reset DENIED
2025-11-13 22:01:24 Edward /reset FAIL
2025-02-08 01:25:51 Ein /admin DENIED
2025-04-21 07:52:14 Ein /admin DENIED
2025-01-01 00:13:16 Ein /download DENIED
2025-10-15 14:35:17 Ein /download SUCCESS
2025-06-08 01:32:46 Ein /logout FAIL
2025-05-12 15:00:19 Eren /404 DENIED
2025-06-15 05:19:04 Eren /download DENIED
2025-06-24 08:20:03 Eren /logout SUCCESS
2025-03-12 15:23:38 Eren /upload FAIL
2025-06-28 16:05:02 Eren /upload FAIL
2025-11-11 03:05:49 Faye /404 DENIED
2025-08-21 21:25:07 Faye /admin SUCCESS
2025-11-27 23:50:38 Faye /dashboard DENIED
2025-09-01 22:22:06 Faye /download DENIED
2025-01-04 22:32:01 Faye /logout FAIL
2025-03-25 21:20:59 Haruka /admin SUCCESS
2025-05-22 13:19:50 Haruka /logout SUCCESS
2025-06-04 02:34:36 Honami /admin DENIED
2025-10-09 15:46:36 Honami /admin SUCCESS
2025-01-22 03:33:11 Honami /logout FAIL
2025-09-18 22:09:07 Honami /logout FAIL
2025-01-18 22:10:43 Jet /download SUCCESS
2025-08-24 15:05:45 Jet /login DENIED
2025-12-04 11:03:31 Jet /logout DENIED
2025-08-04 18:36:17 Jet /reset DENIED
2025-12-22 13:13:44 Jet /reset FAIL
2025-12-18 03:33:07 Jet /settings FAIL
2025-08-26 17:10:09 Jet /settings SUCCESS
2025-11-09 09:19:01 Julia /404 DENIED
```

```
frank@cis106:~$ sort -k4 -o sorted_access_by_endpoint.log access.log
frank@cis106:~$ cat sorted_access_by_endpoint.log
2025-05-12 15:00:19 Eren /404 DENIED
2025-05-26 20:43:46 Edward /404 DENIED
2025-11-09 09:19:01 Julia /404 DENIED
2025-11-11 03:05:49 Faye /404 DENIED
2025-12-01 06:43:14 Arisu /404 FAIL
2025-06-20 20:11:37 Seras /404 SUCCESS
2025-07-11 07:04:18 Edward /404 SUCCESS
2025-09-10 15:35:59 Rei /404 SUCCESS
2025-12-08 21:06:19 Edward /404 SUCCESS
2025-01-12 05:45:12 Shinji /admin DENIED
2025-01-26 21:22:50 Rize /admin DENIED
2025-02-08 01:25:51 Ein /admin DENIED
2025-02-25 11:05:38 Rize /admin DENIED
2025-04-21 07:52:14 Ein /admin DENIED
2025-06-04 02:34:36 Honami /admin DENIED
2025-04-23 16:03:34 Touka /admin FAIL
2025-03-25 21:20:59 Haruka /admin SUCCESS
2025-05-21 16:53:20 Touka /admin SUCCESS
2025-06-24 02:13:00 Julia /admin SUCCESS
2025-08-21 21:25:07 Faye /admin SUCCESS
2025-08-27 14:07:23 Edward /admin SUCCESS
2025-09-15 09:51:26 Annie /admin SUCCESS
2025-10-09 15:46:36 Honami /admin SUCCESS
2025-12-22 21:26:24 Walter /admin SUCCESS
2025-01-01 06:02:31 Seras /dashboard DENIED
2025-08-19 14:29:32 Julia /dashboard DENIED
2025-11-12 07:37:12 Ken /dashboard DENIED
2025-11-27 23:50:38 Faye /dashboard DENIED
2025-01-03 17:09:02 Ken /dashboard FAIL
2025-10-04 01:57:16 Walter /dashboard FAIL
2025-03-08 21:10:30 Rize /dashboard SUCCESS
2025-06-20 04:28:48 Annie /dashboard SUCCESS
2025-12-05 16:20:59 Edward /dashboard SUCCESS
2025-01-01 00:13:16 Ein /download DENIED
2025-04-24 20:42:52 Julia /download DENIED
2025-06-15 05:19:04 Eren /download DENIED
2025-06-25 23:54:08 Arisu /download DENIED
2025-09-01 22:22:06 Faye /download DENIED
2025-03-08 13:07:23 Rei /download FAIL
2025-03-26 02:45:52 Seras /download FAIL
2025-04-08 20:51:04 Kei /download FAIL
2025-08-23 20:02:11 Touka /download FAIL
2025-10-10 11:57:04 Touka /download FAIL
2025-12-14 21:48:13 Ken /download FAIL
2025-01-18 22:10:43 Jet /download SUCCESS
2025-04-17 00:27:50 Annie /download SUCCESS
```

```
Rei /download
frank@cis106:~$ cut -d" " -f3,4 --output-delimiter=" accessed: " access.log
Ken accessed: /dashboard
Edward accessed: /reset
Jet accessed: /settings
Honami accessed: /admin
Rei accessed: /settings
Asimov accessed: /logout
Asimov accessed: /upload
Edward accessed: /reset
Touka accessed: /download
Rize accessed: /admin
Honami accessed: /logout
Jet accessed: /logout
Shinji accessed: /upload
Kei accessed: /reset
Shinji accessed: /login
Kei accessed: /settings
Touka accessed: /download
Arisu accessed: /reset
Kei accessed: /reset
Jet accessed: /login
Rei accessed: /reset
Ein accessed: /logout
Ein accessed: /download
Julia accessed: /dashboard
Seras accessed: /404
Touka accessed: /login
Seras accessed: /reset
Asimov accessed: /logout
Walter accessed: /admin
Ein accessed: /download
Eren accessed: /404
Faye accessed: /dashboard
Seras accessed: /dashboard
Ken accessed: /dashboard
Edward accessed: /admin
Edward accessed: /404
Ken accessed: /logout
Asimov accessed: /reset
Ein accessed: /admin
Faye accessed: /404
Edward accessed: /404
Annie accessed: /download
Annie accessed: /admin
Edward accessed: /dashboard
Ken accessed: /settings
Ein accessed: /admin
```